POWER BI - WEEK 11 – REPORT

INTRODUCTION

Data cleaning and transformation are important steps in preparing raw automobile data for analysis. In this activity, the Automobile Dataset is imported into Power BI and processed to improve data quality and prepare it for automobile analytics.

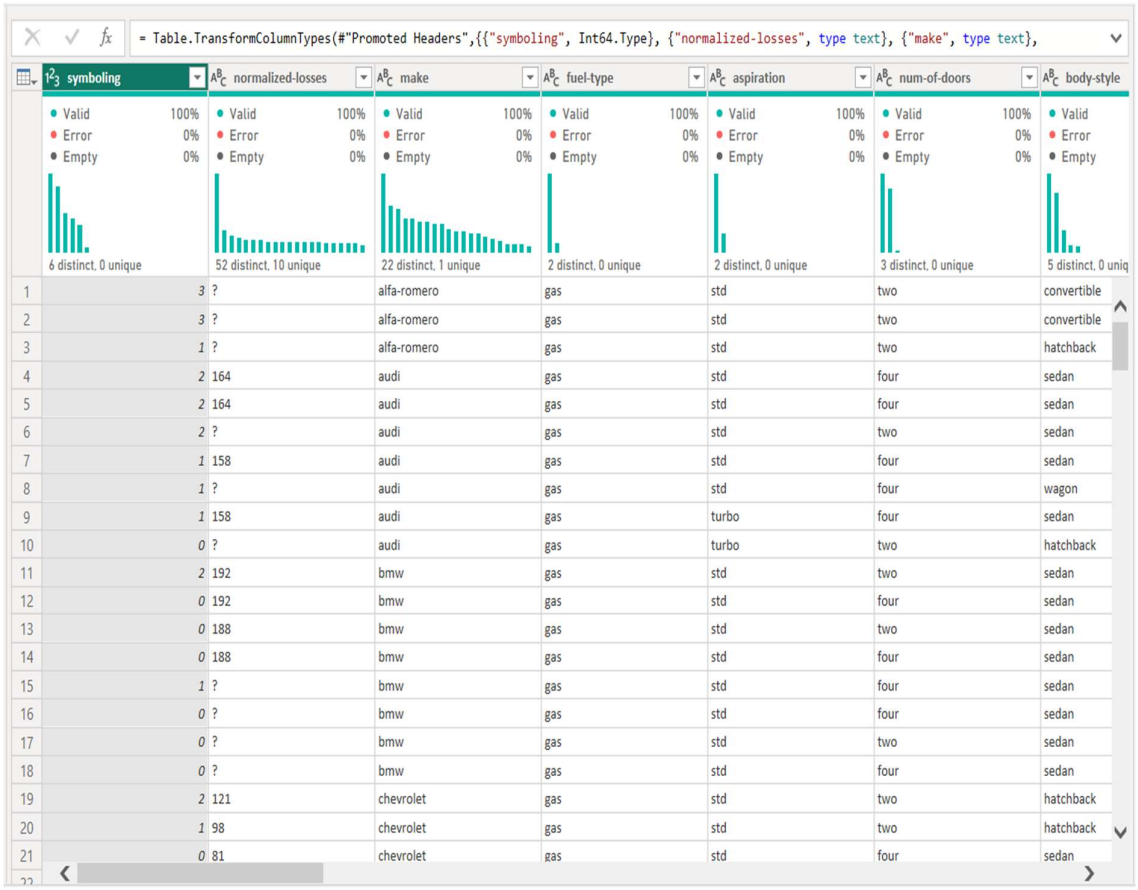
The activity includes handling invalid symbol placeholders, correcting data types, managing missing values, creating calculated columns, grouping vehicles into meaningful categories, and developing DAX measures. The cleaned dataset will be used to create an interactive Automobile Analytics Dashboard in Week 12.

Name: Durga Devi S

Roll No: E25DA018

Department: Data Science and Artificial Intelligence(||)

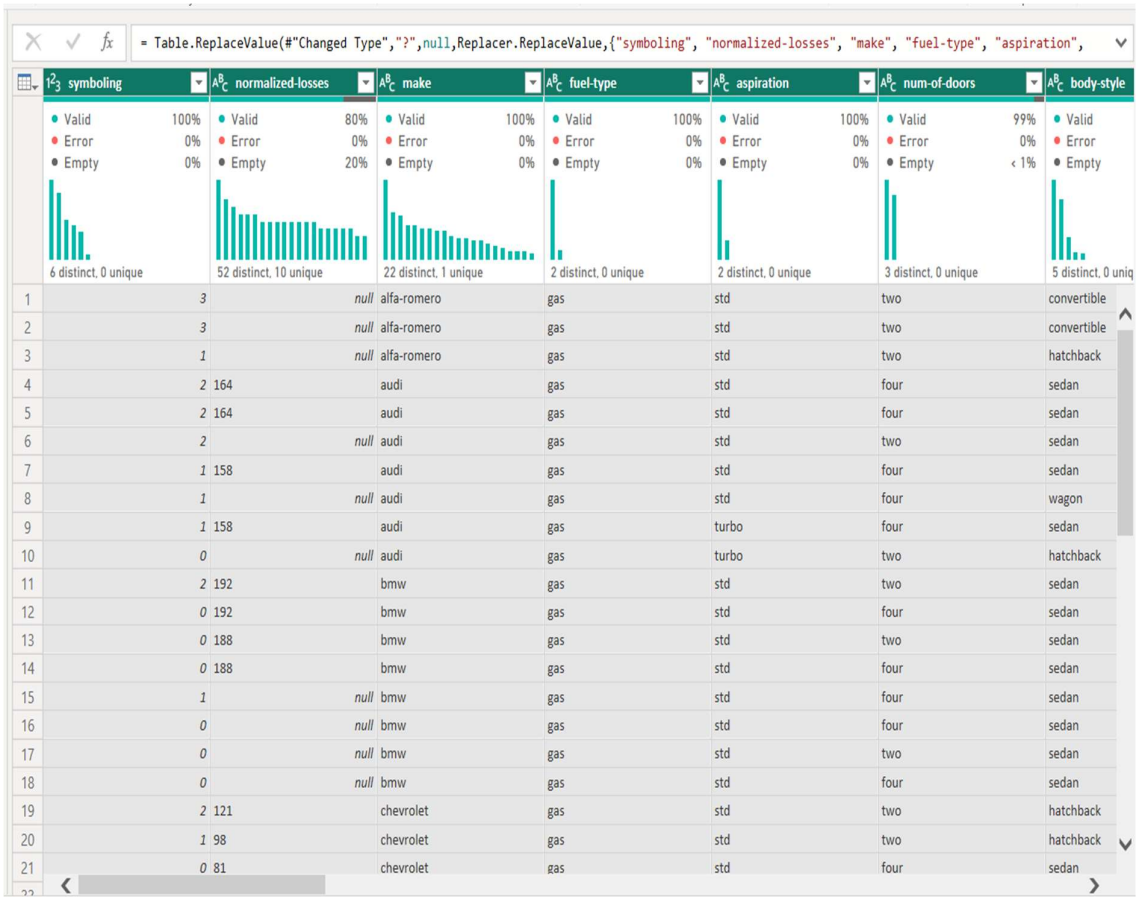
Figure 1: Importing and Inspecting the Automobile Dataset in Power BI



= Table.TransformColumnTypes(#"Promoted Headers",{{"symboling", Int64.Type}, {"normalized-losses", type text}, {"make", type text},

	symboling	normalized-losses	make	fuel-type	aspiration	num-of-doors	body-style
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
	6 distinct, 0 unique	52 distinct, 10 unique	22 distinct, 1 unique	2 distinct, 0 unique	2 distinct, 0 unique	3 distinct, 0 unique	5 distinct, 0 uniq
1	3	?	alfa-romero	gas	std	two	convertible
2	3	?	alfa-romero	gas	std	two	convertible
3	1	?	alfa-romero	gas	std	two	hatchback
4	2	164	audi	gas	std	four	sedan
5	2	164	audi	gas	std	four	sedan
6	2	?	audi	gas	std	two	sedan
7	1	158	audi	gas	std	four	sedan
8	1	?	audi	gas	std	four	wagon
9	1	158	audi	gas	turbo	four	sedan
10	0	?	audi	gas	turbo	two	hatchback
11	2	192	bmw	gas	std	two	sedan
12	0	192	bmw	gas	std	four	sedan
13	0	188	bmw	gas	std	two	sedan
14	0	188	bmw	gas	std	four	sedan
15	1	?	bmw	gas	std	four	sedan
16	0	?	bmw	gas	std	four	sedan
17	0	?	bmw	gas	std	two	sedan
18	0	?	bmw	gas	std	four	sedan
19	2	121	chevrolet	gas	std	two	hatchback
20	1	98	chevrolet	gas	std	two	hatchback
21	0	81	chevrolet	gas	std	four	sedan

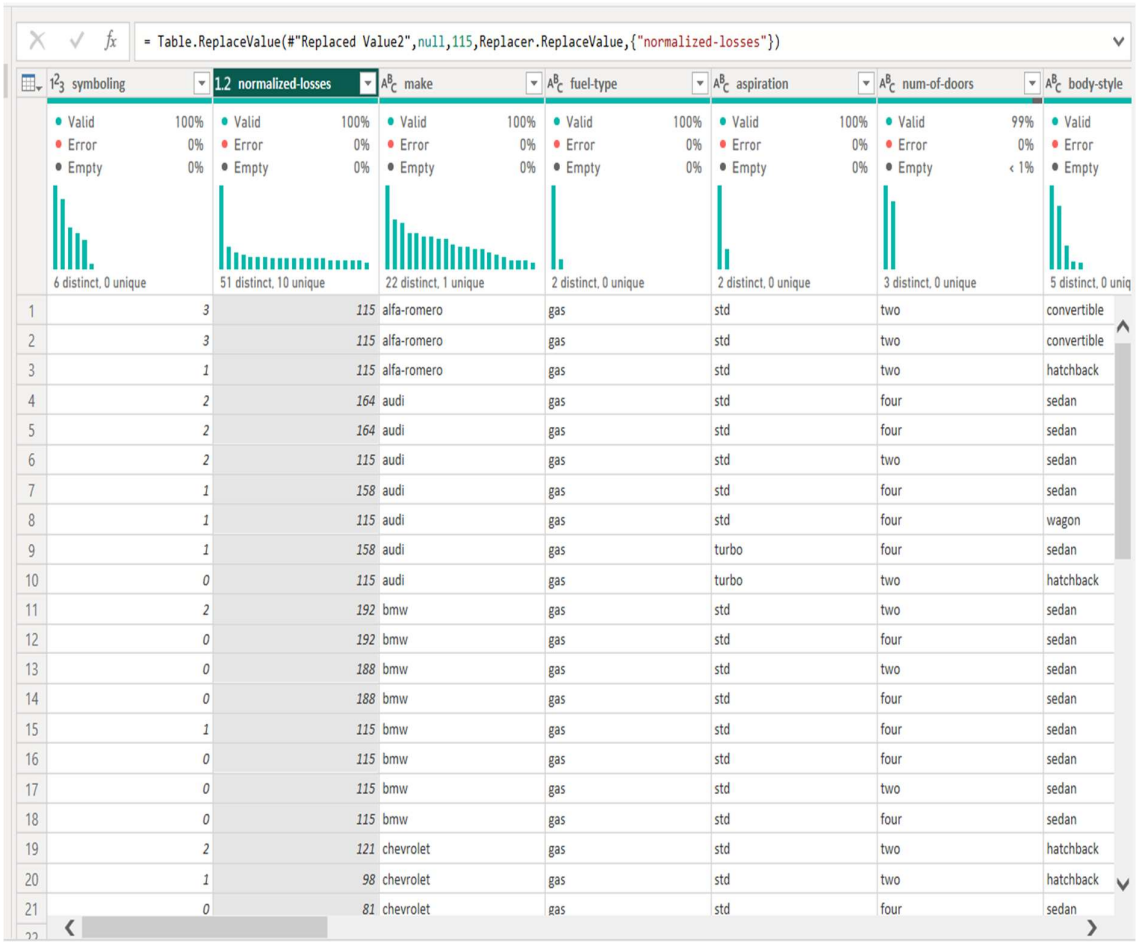
Figure 2: Replacing Invalid Symbol Placeholders with Null Values in Power Query



```
= Table.ReplaceValue(#"Changed Type","?",null,Replacer.ReplaceValue,{"symboling", "normalized-losses", "make", "fuel-type", "aspiration",
```

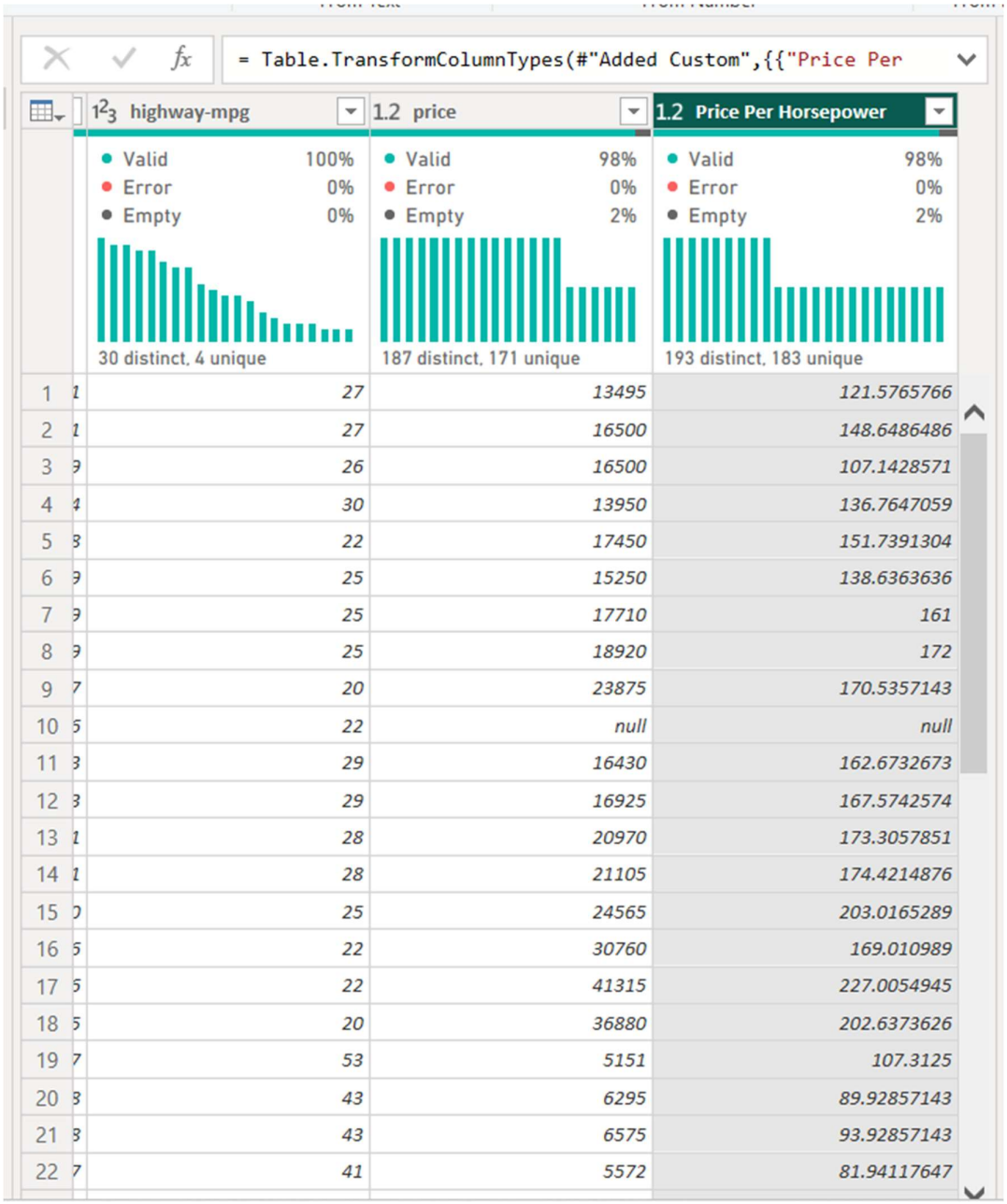
	symboling	normalized-losses	make	fuel-type	aspiration	num-of-doors	body-style
	Valid 100% Error 0% Empty 0% 6 distinct, 0 unique	Valid 80% Error 0% Empty 20% 52 distinct, 10 unique	Valid 100% Error 0% Empty 0% 22 distinct, 1 unique	Valid 100% Error 0% Empty 0% 2 distinct, 0 unique	Valid 100% Error 0% Empty 0% 2 distinct, 0 unique	Valid 99% Error 0% Empty < 1% 3 distinct, 0 unique	Valid Error Empty 5 distinct, 0 uniq
1	3	null	alfa-romero	gas	std	two	convertible
2	3	null	alfa-romero	gas	std	two	convertible
3	1	null	alfa-romero	gas	std	two	hatchback
4	2	164	audi	gas	std	four	sedan
5	2	164	audi	gas	std	four	sedan
6	2	null	audi	gas	std	two	sedan
7	1	158	audi	gas	std	four	sedan
8	1	null	audi	gas	std	four	wagon
9	1	158	audi	gas	turbo	four	sedan
10	0	null	audi	gas	turbo	two	hatchback
11	2	192	bmw	gas	std	two	sedan
12	0	192	bmw	gas	std	four	sedan
13	0	188	bmw	gas	std	two	sedan
14	0	188	bmw	gas	std	four	sedan
15	1	null	bmw	gas	std	four	sedan
16	0	null	bmw	gas	std	four	sedan
17	0	null	bmw	gas	std	two	sedan
18	0	null	bmw	gas	std	four	sedan
19	2	121	chevrolet	gas	std	two	hatchback
20	1	98	chevrolet	gas	std	two	hatchback
21	0	81	chevrolet	gas	std	four	sedan

Figure 3: Correcting Technical Data Types and Handling Missing Values in Power Query



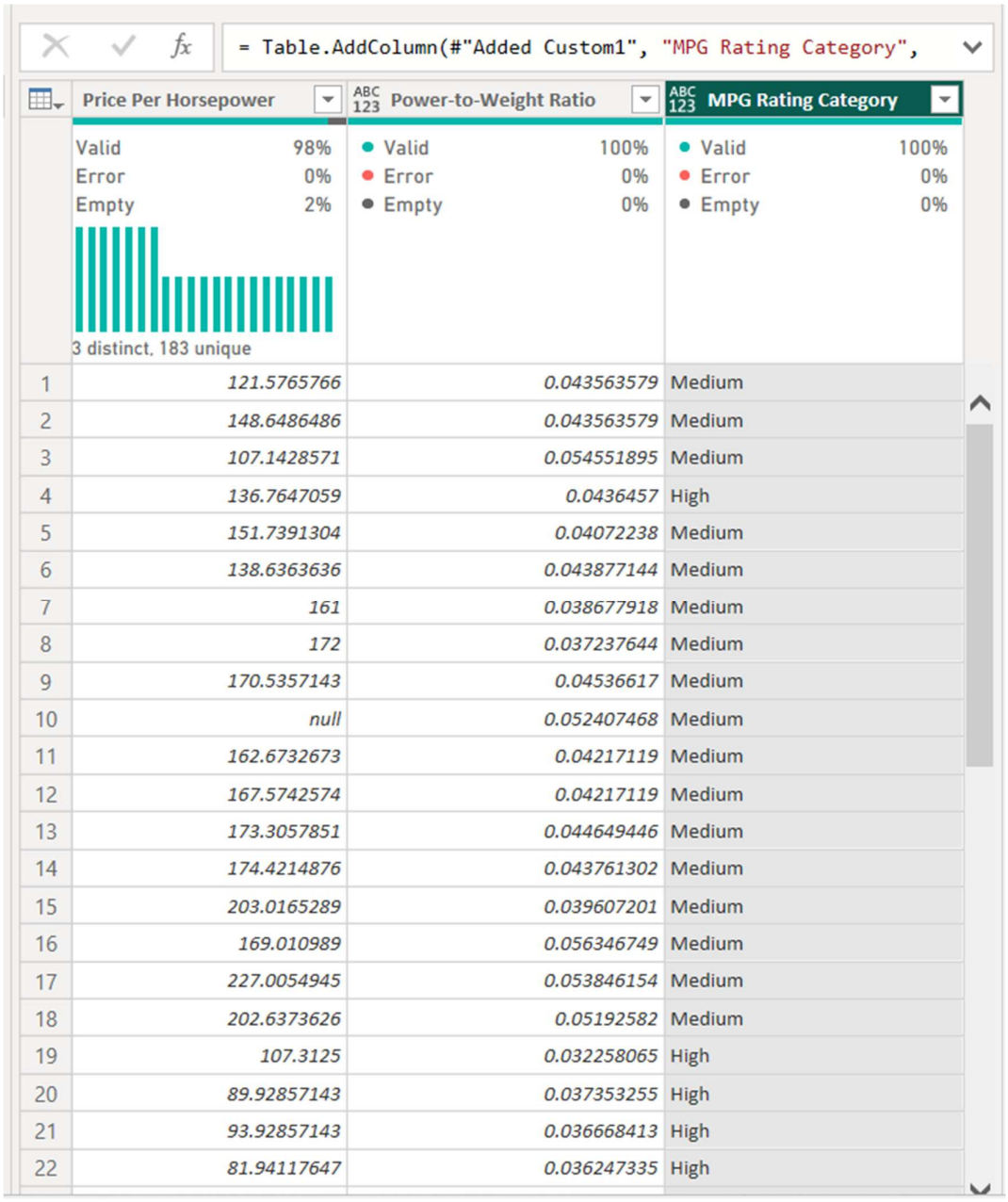
	symboling	normalized-losses	make	fuel-type	aspiration	num-of-doors	body-style
1	3	115	alfa-romero	gas	std	two	convertible
2	3	115	alfa-romero	gas	std	two	convertible
3	1	115	alfa-romero	gas	std	two	hatchback
4	2	164	audi	gas	std	four	sedan
5	2	164	audi	gas	std	four	sedan
6	2	115	audi	gas	std	two	sedan
7	1	158	audi	gas	std	four	sedan
8	1	115	audi	gas	std	four	wagon
9	1	158	audi	gas	turbo	four	sedan
10	0	115	audi	gas	turbo	two	hatchback
11	2	192	bmw	gas	std	two	sedan
12	0	192	bmw	gas	std	four	sedan
13	0	188	bmw	gas	std	two	sedan
14	0	188	bmw	gas	std	four	sedan
15	1	115	bmw	gas	std	four	sedan
16	0	115	bmw	gas	std	four	sedan
17	0	115	bmw	gas	std	two	sedan
18	0	115	bmw	gas	std	four	sedan
19	2	121	chevrolet	gas	std	two	hatchback
20	1	98	chevrolet	gas	std	two	hatchback
21	0	81	chevrolet	gas	std	four	sedan

Figure 4: Creating the Price per Horsepower Calculated Column in Power Query



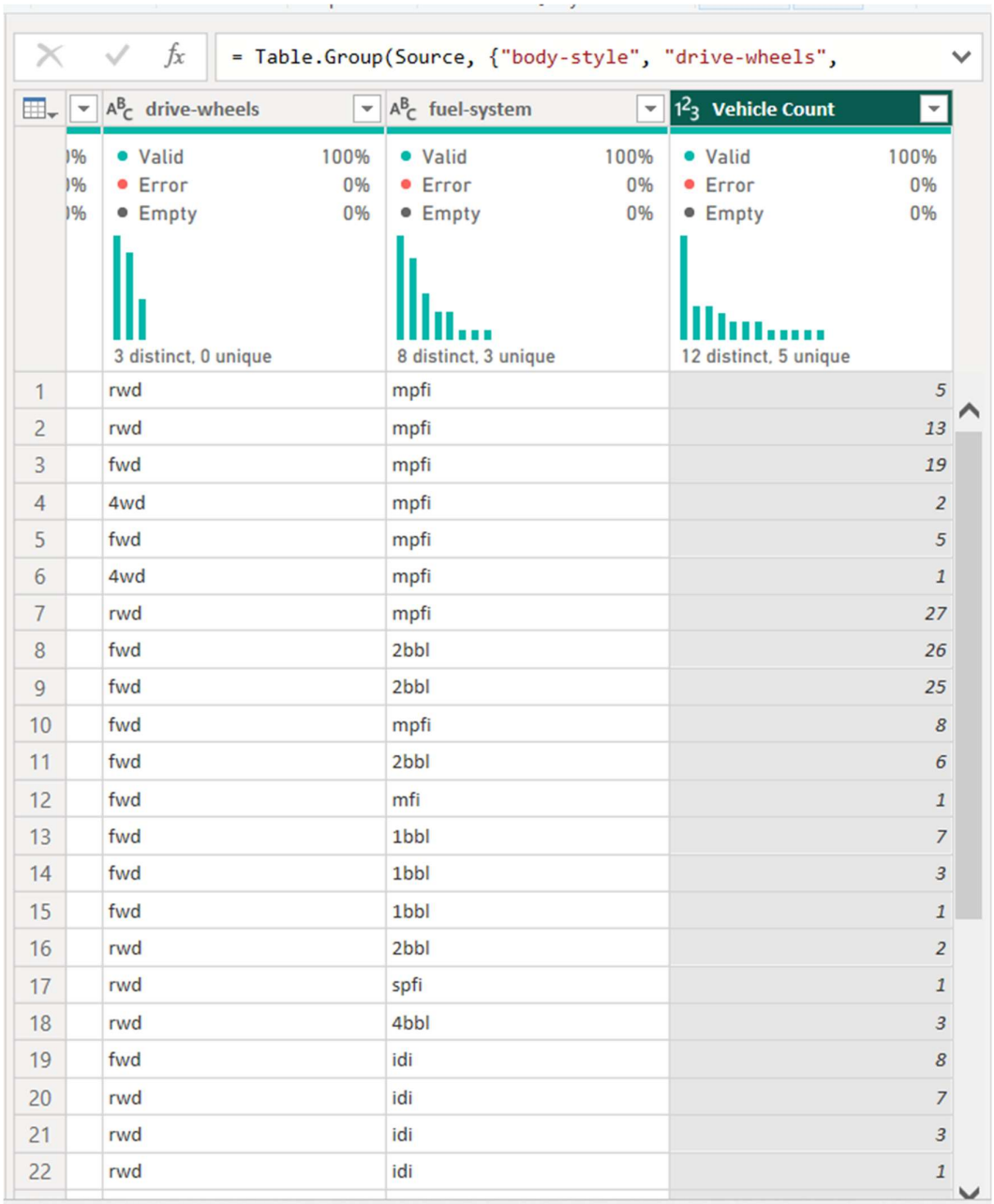
	highway-mpg	price	Price Per Horsepower
1	27	13495	121.5765766
2	27	16500	148.6486486
3	26	16500	107.1428571
4	30	13950	136.7647059
5	22	17450	151.7391304
6	25	15250	138.6363636
7	25	17710	161
8	25	18920	172
9	20	23875	170.5357143
10	22	null	null
11	29	16430	162.6732673
12	29	16925	167.5742574
13	28	20970	173.3057851
14	28	21105	174.4214876
15	25	24565	203.0165289
16	22	30760	169.010989
17	22	41315	227.0054945
18	20	36880	202.6373626
19	53	5151	107.3125
20	43	6295	89.92857143
21	43	6575	93.92857143
22	41	5572	81.94117647

Figure 5: Creating Power-to-Weight Ratio and MPG Rating Category in Power Query



	Price Per Horsepower	Power-to-Weight Ratio	MPG Rating Category
1	121.5765766	0.043563579	Medium
2	148.6486486	0.043563579	Medium
3	107.1428571	0.054551895	Medium
4	136.7647059	0.0436457	High
5	151.7391304	0.04072238	Medium
6	138.6363636	0.043877144	Medium
7	161	0.038677918	Medium
8	172	0.037237644	Medium
9	170.5357143	0.04536617	Medium
10	null	0.052407468	Medium
11	162.6732673	0.04217119	Medium
12	167.5742574	0.04217119	Medium
13	173.3057851	0.044649446	Medium
14	174.4214876	0.043761302	Medium
15	203.0165289	0.039607201	Medium
16	169.010989	0.056346749	Medium
17	227.0054945	0.053846154	Medium
18	202.6373626	0.05192582	Medium
19	107.3125	0.032258065	High
20	89.92857143	0.037353255	High
21	93.92857143	0.036668413	High
22	81.94117647	0.036247335	High

Figure 6: Grouping Automobile Data by Body Style, Drive Wheels, and Fuel System in Power BI



	drive-wheels	fuel-system	Vehicle Count
1	rwd	mpfi	5
2	rwd	mpfi	13
3	fwd	mpfi	19
4	4wd	mpfi	2
5	fwd	mpfi	5
6	4wd	mpfi	1
7	rwd	mpfi	27
8	fwd	2bbl	26
9	fwd	2bbl	25
10	fwd	mpfi	8
11	fwd	2bbl	6
12	fwd	mfi	1
13	fwd	1bbl	7
14	fwd	1bbl	3
15	fwd	1bbl	1
16	rwd	2bbl	2
17	rwd	spfi	1
18	rwd	4bbl	3
19	fwd	idi	8
20	rwd	idi	7
21	rwd	idi	3
22	rwd	idi	1

Figure 7: Creating DAX Measures for Average Price and Average Horsepower in Power BI

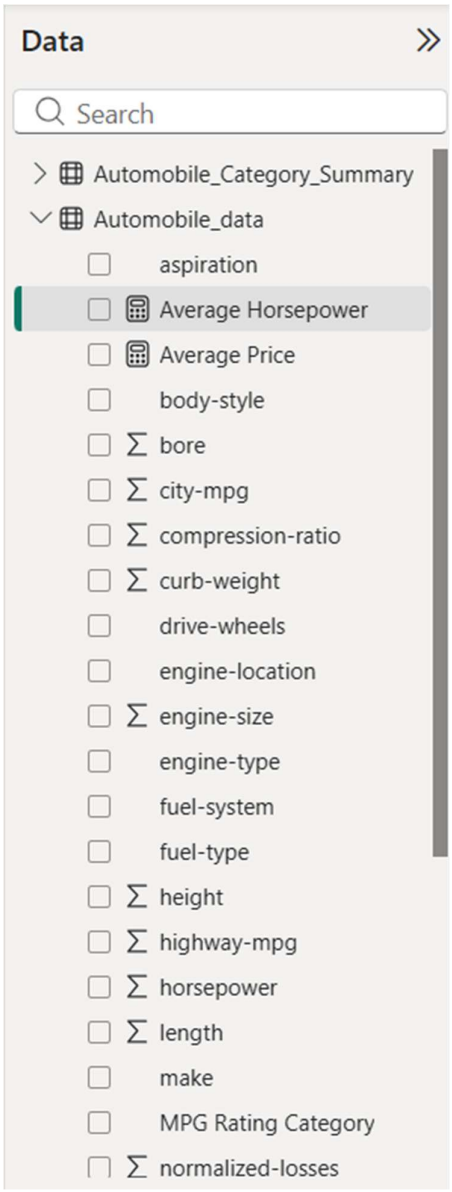


Figure 8: Creating DAX Measures for Average Engine Size and Average Highway MPG in Power BI

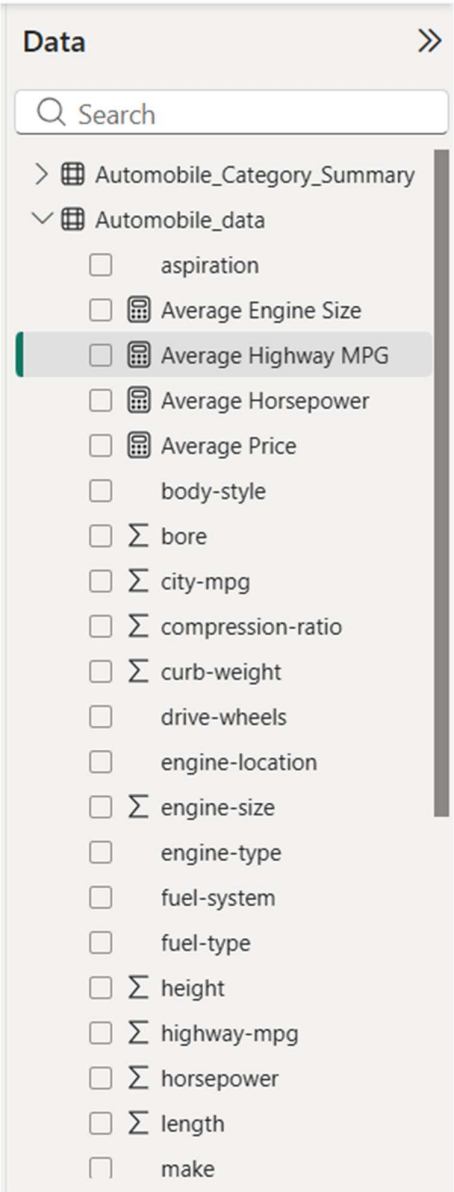
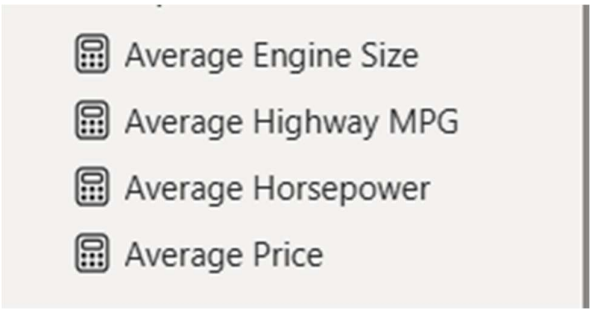
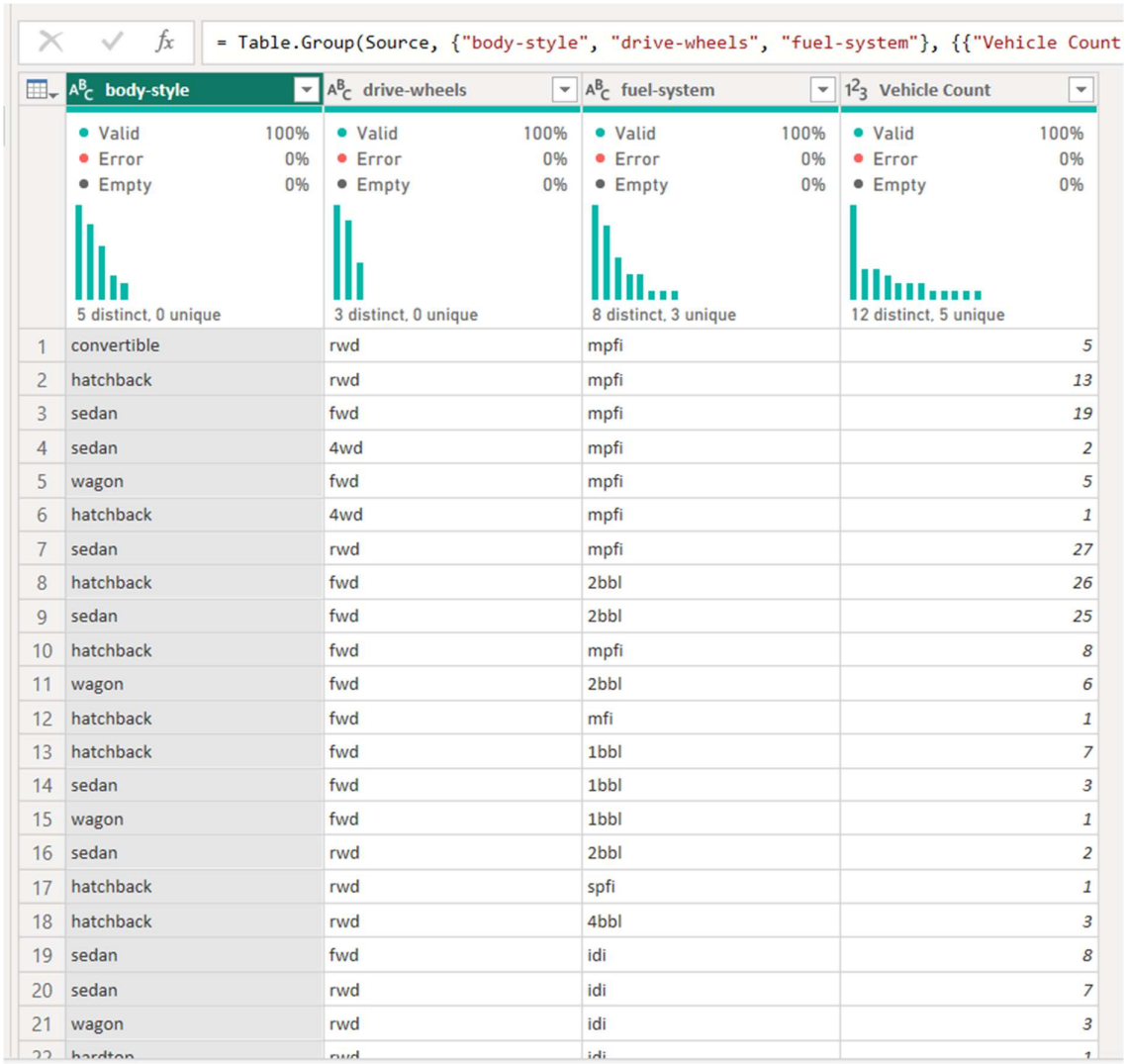


Figure 9: Final Validating Cleaned Automobile Data, Calculated Columns, and DAX Measures in Power BI

Price Per Horsepower	Power-to-Weight Ratio	MPG Rating Category
121.576576576577	0.043563579	Medium
148.648648648649	0.043563579	Medium
107.142857142857	0.054551895	Medium
136.764705882353	0.0436457	High
151.739130434783	0.04072238	Medium
138.636363636364	0.043877144	Medium
161	0.038677918	Medium
172	0.037237644	Medium
170.535714285714	0.04536617	Medium
	0.052407468	Medium
162.673267326733	0.04217119	Medium
167.574257425743	0.04217119	Medium
173.305785123967	0.044649446	Medium
174.421487603306	0.043761302	Medium
203.01652892562	0.039607201	Medium
169.010989010989	0.056346749	Medium
227.005494505495	0.053846154	Medium
202.637362637363	0.05192582	Medium
107.3125	0.032258065	High
89.9285714285714	0.037353255	High
93.9285714285714	0.036668413	High
81.9411764705882	0.036247335	High
93.7794117647059	0.036247335	High
78.0098039215686	0.047932331	High
91.6029411764706	0.034570412	High
98.4117647058823	0.034188034	High
111.897058823529	0.034188034	High





	body-style	drive-wheels	fuel-system	Vehicle Count
1	convertible	rwd	mpfi	5
2	hatchback	rwd	mpfi	13
3	sedan	fwd	mpfi	19
4	sedan	4wd	mpfi	2
5	wagon	fwd	mpfi	5
6	hatchback	4wd	mpfi	1
7	sedan	rwd	mpfi	27
8	hatchback	fwd	2bbl	26
9	sedan	fwd	2bbl	25
10	hatchback	fwd	mpfi	8
11	wagon	fwd	2bbl	6
12	hatchback	fwd	mfi	1
13	hatchback	fwd	1bbl	7
14	sedan	fwd	1bbl	3
15	wagon	fwd	1bbl	1
16	sedan	rwd	2bbl	2
17	hatchback	rwd	spfi	1
18	hatchback	rwd	4bbl	3
19	sedan	fwd	idi	8
20	sedan	rwd	idi	7
21	wagon	rwd	idi	3
22	hardtop	rwd	idi	1

CONCLUSION

The Week 11 activity successfully completed automobile data cleaning and transformation using Power BI. Invalid placeholders were replaced with null values, missing numeric values were handled using median imputation, and appropriate data types were assigned.

New calculated columns were created for Price per Horsepower, Power-to-Weight Ratio, and MPG Rating Category. Automobile records were grouped by body style, drive wheels, and fuel system. DAX measures were also created to calculate Average Price, Average Horsepower, Average Engine Size, and Average Highway MPG.

This activity provided practical experience in data preprocessing, data transformation, calculated columns, grouping, and DAX calculations, preparing the dataset for further automobile analysis.